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Data Engineering Roadmap

1. Overview

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This roadmap is designed to build strong foundations in Data Engineering, covering core concepts, tools, and real-world project implementation.

Focus Areas

- Core Concepts
- Programming and Data Processing
- Cloud and Big Data Technologies
- Real-Time Project Implementation

2. Learning Roadmap

Phase 1: Foundations of Data Engineering

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2.1 Data Engineering Fundamentals

-  Engineering
- Roles and responsibilities of a Data Engineer
- Data Engineer vs Data Analyst vs Data Scientist
- Data lifecycle:
 - Data ingestion
 - Data processing
 - Data storage
 - Data consumption
- Types of data:
 - Structured
 - Semi-structured
 - Unstructured

2.2 Data Pipeline Concepts

- **ETL vs ELT**
- **Batch processing vs Real-time processing**
- **Data pipeline architecture**
- **Data flow design patterns**
- **Introduction to tools:**
 - **Apache Kafka**
 - **Apache Spark**
 - **Apache Airflow**
- **Data Lake vs Data Warehouse**

2.3 Databases

SQL Databases

- **RDBMS concepts**
- **ACID properties**
- **Schema design basics**
- **Normalization**

NoSQL Databases

- **Types:**
 - **Key-Value**
 - **Document**
 - **Column**
 - **Graph**
- **Use cases of SQL vs NoSQL**

2.4 Linux Fundamentals

- **File system structure:**
 - **/home, /etc, /var, /tmp**
- **File operations:**
 - **ls, cd, pwd, cp, mv, rm, touch**
- **File viewing:**

- **cat, less, head, tail**
- **Searching and processing:**
 - **grep, find, awk, sed**
- **File permissions and ownership**
- **Process management:**
 - **ps, top, kill**
- **Job scheduling:**
 - **cron**
- **Shell scripting basics:**
 - **variables, loops, conditions**

2.5 SQL Fundamentals

- **SQL execution order**
- **Basic queries:**
 - **SELECT, WHERE, ORDER BY**
- **Aggregations:**
 - **GROUP BY, HAVING**
- **Joins:**
 - **INNER, LEFT, RIGHT, FULL**
- **Functions:**
 - **COUNT, SUM, AVG, MIN, MAX**
- **Data manipulation:**
 - **INSERT, UPDATE, DELETE**
- **Constraints:**
 - **PRIMARY KEY, FOREIGN KEY, UNIQUE**
- **Index basics**

3.1 Python Fundamentals

- **Syntax and structure**
- **Data types:**
 - int, float, string, boolean
- **Data structures:**
 - list, tuple, set, dictionary
- **Control flow:**
 - if-else, loops
- **Functions and modules**
- **Exception handling**

3.2 Python for Data Engineering

- **File handling:**
 - CSV, JSON, TXT
- **Data processing scripts**
- **Working with APIs (requests)**
- **Logging and error handling**
- **Virtual environments**

3.3 Python Libraries



pandas

- **DataFrames**
- **Filtering and selection**
- **Grouping and aggregation**
- **Merging and joining**

numpy

- **Arrays**
- **Numerical operations**

3.4 PySpark Fundamentals

- **Spark architecture**
- **RDD vs DataFrame vs Dataset**
- **SparkSession**
- **Transformations:**
 - **map, flatMap, filter**
- **Actions:**
 - **collect, count, show**
- **Reading and writing data:**
 - **CSV, JSON, Parquet**

3.5 Advanced PySpark

- **Joins and aggregations**
- **Window functions**
- **Partitioning:**
 - **repartition, coalesce**
- **Performance optimization:**
 - **caching, broadcasting**

Phase 3: Cloud and Data Engineering Tools

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4.1 Cloud Fundamentals

- **Cloud computing concepts:**
 - **IaaS, PaaS, SaaS**
- **Benefits of cloud computing**

4.2 Microsoft Azure

- **Azure Blob Storage**
- **Azure Data Lake**
- **Azure SQL Database**

- **Azure Databricks**
 - **Azure Data Factory**
 - **Resource management:**
 - **Azure Portal**
 - **Azure CLI**
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Phase 4: Advanced Data Engineering

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5.1 Advanced SQL

- **Subqueries (correlated and non-correlated)**
- **Common Table Expressions (CTEs)**
- **Window functions:**
 - **ROW_NUMBER, RANK, DENSE_RANK**
 - **LEAD, LAG**
- **Case statements**
- **Query optimization**
- **Indexing strategies**
- **Partitioning**

5.2 Data Warehousing

- **OLTP vs OLAP**
- **Star schema**
- **Snowflake schema**
- **Fact and dimension tables**
- **Slowly Changing Dimensions (SCD)**
- **Data marts**

5.3 Data Pipelines

- **ETL pipeline design**
- **Data ingestion:**
 - **APIs**
 - **Files**
 - **Databases**

- **Data transformation**
- **Data validation**

5.4 Workflow Orchestration

- **Apache Airflow:**
 - **DAGs**
 - **Task scheduling**
- **Azure Data Factory:**
 - **Pipelines**
 - **Linked services**

5.5 Best Practices

- **Data quality checks**
- **Handling missing data**
- **Logging and monitoring**
- **Version control using Git**
- **Testing:**
 - **Unit testing**
 - **Integration testing**

Phase 5: Real-Time Project



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6.1 Project Scope

- **Data ingestion:**
 - **APIs**
 - **Databases**
 - **Files**
- **Data transformation:**
 - **Python**
 - **PySpark**
- **Pipeline development:**
 - **Batch or real-time**

- **Data storage:**

- **Azure Blob**
- **Data Lake**

6.2 Advanced Components

- **Streaming (optional):**

- **Apache Kafka**
- **Spark Streaming**

6.3 Automation

- **Apache Airflow**
- **Azure Data Factory**

6.4 Deployment

- **Cloud deployment**
- **Monitoring and logging**
- **Performance optimization**

6.5 Deliverables

- **Source code (Git repository)**
- **Documentation**
- **Architecture diagram**
- **Final presentation**



3. Tools and Technologies

Languages

- **SQL**
- **Python**

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Big Data

- **Apache Spark (PySpark)**
- **Apache Kafka**

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Cloud

- **Microsoft Azure**

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Orchestration

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- Apache Airflow
- Azure Data Factory

Databases

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- PostgreSQL
- MySQL
- MongoDB

4. Final Guidance

- Focus on hands-on practice
 - Build real-world data pipelines
 - Keep concepts clear and structured
 - Practice debugging and optimization
 - Maintain proper documentation for all work
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